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The Partition of Cosmological Tensions: Frame Artifacts, Local Structure, and Physical Residuals

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ABSTRACT

We apply ξ-normalized coordinates ($\xi \equiv d_c/d_H$, where $d_H = c/H_0$) to partition ten cosmological tensions into three distinct classes: (1) coordinate-frame artifacts arising from H_0-dependent measurement comparisons, (2) local-structure effects including the KBC void and bulk flows, and (3) genuine physical residuals requiring new cosmological physics. Across the full tension landscape — Hubble constant, matter clustering (S_8), baryon acoustic oscillations, lensing amplitude (A_L), age, and Lyman-α forest — we find that ∼90% of the H_0 tension is a frame artifact, 64% of the local-expansion anomaly is explained by the KBC void and bulk flow, and the strong-lensing H0LiCOW tension is fully accounted for by the mass-sheet degeneracy. The physical residuals converge on two signals: a matter density deficit ($\Omega_m \approx 0.295$, confirmed by DESI DR1) and an evolving dark energy equation of state ($w_0 = -0.634$, $w_a = -1.388$, $\Delta\chi^2 = 4.74$). A multi-probe convergence test demonstrates that the DESI low-z BAO w_0w_a fit extrapolates to the Ly-α BAO at $z = 2.33$ within 0.94σ, suggesting a coherent dark energy evolution signal across $z = 0.3$–2.3. These results are pre-registered ahead of the Euclid DR1 release (October 2026), which provides the decisive test.

Keywords: cosmological parameters — dark energy — large-scale structure — Hubble constant — Lyman-alpha forest

1. INTRODUCTION

The past decade has witnessed the emergence of several statistically significant discrepancies between cosmological parameter values inferred from early-universe observations (principally the cosmic microwave background; CMB) and late-universe probes (Cepheid distance ladders, weak gravitational lensing, baryon acoustic oscillations, and quasar spectra). The most prominent of these is the "Hubble tension": a 4–5σ discrepancy between the locally measured expansion rate ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}; Riess et al. 2022) and the CMB-inferred value ($H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}; Planck Collaboration 2020).

The dominant response to these tensions has been to propose modifications to the standard ΛCDM model — early dark energy, interacting dark matter, modified gravity — or to search for unidentified systematic errors in one or more datasets (Di Valentino et al. 2021). A complementary possibility, which we investigate here, is that a significant fraction of the apparent tension is a *coordinate-frame artifact*: a consequence of comparing measurements that implicitly assume different values of H_0 in their distance definitions.

In Martin (2026) (hereafter Paper I), we demonstrated that the redshift-scaled comoving distance $\xi \equiv d_c/d_H = d_cH_0/c$ provides a frame-independent coordinate in which H_0-dependent systematics cancel to first order. Applied to the SH0ES Cepheid calibrator sample, ξ-normalization reduces the apparent H_0 discrepancy by ∼93%, recovering a physical residual of ∼7% attributable to a matter density deficit rather than an expansion rate anomaly.

Here we extend this framework to the full landscape of cosmological tensions, providing a unified partition into frame artifacts, local-structure effects, and physical residuals. Our goal is not to resolve the tensions by fiat, but to identify precisely *where* new physics is required and where it is not.

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2. METHODS

2.1. *The ξ Normalization*

The ξ-coordinate is defined as

$$\xi(z) \equiv \frac{d_c(z)}{d_H} = \frac{H_0}{c}\int_0^z \frac{dz'}{E(z')}, \tag{1}$$

where $E(z) = H(z)/H_0$ is the dimensionless Hubble parameter. For a flat w_0w_a cosmology (Chevallier & Polarski 2001; Linder 2003),

$$E(z) = \left[\Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0+w_a)}e^{-3w_a z/(1+z)}\right]^{1/2}. \tag{2}$$

Because ξ is dimensionless and H_0-independent, comparisons of ξ values between datasets with different assumed H_0 do not introduce frame-mixing artifacts. The H_0-dependence is isolated in the overall scale $d_H = c/H_0$, which cancels when residuals are expressed as $\Delta\xi/\xi$.

2.2. *Frame-Artifact Decomposition*

For each tension, we compute: (1) the raw tension in conventional coordinates (km s^{-1} Mpc^{-1} or dimensionless parameter ratios); (2) the tension after ξ-normalization with a frame-independent anchor $H_{0,\mathrm{ref}} = 70.0$ km s^{-1} Mpc^{-1}; and (3) the physical residual, defined as the tension that persists after both ξ-normalization and (where applicable) corrections for local structure.

The frame-artifact fraction is estimated as

$$f_{\mathrm{frame}} = 1 - \frac{|\Delta_\xi|}{|\Delta_{\mathrm{raw}}|}, \tag{3}$$

where Δ_{raw} is the raw parameter discrepancy and Δ_ξ is the discrepancy after ξ-normalization.

2.3. *Data*

We use the following published datasets:

- DESI DR1 BAO: D_M/r_s and D_H/r_s at 7 redshift bins, $z = 0.295$–2.33 (DESI Collaboration 2025)
- SH0ES Cepheid calibrators (77 hosts) from Pantheon+SH0ES (Riess et al. 2022)
- Planck 2020 CMB: H_0, Ω_m, σ_8 (Planck Collaboration 2020)
- KiDS-1000, DES-Y3, HSC-Y3 S_8 measurements (Asgari et al. 2021; Amon et al. 2022; Dalal et al. 2023)
- H0LiCOW 6-lens, TDCOSMO 7-lens, TDCOSMO+SLACS (Wong et al. 2020; Millon et al. 2020; Birrer et al. 2021)
- eBOSS Ly-α BAO at $z = 2.33$ (du Mas des Bourboux et al. 2020)
- KBC void density contrast and bulk flow (Keenan et al. 2013; Kenworthy et al. 2019; Watkins et al. 2023)

All analyses use published summary statistics; no raw catalogs are required. Scripts are archived at https://zenodo.org/records/19230366 and the companion repository https://github.com/abba-01/uha.

3. RESULTS

3.1. *The Tension Partition Table*

Table 1 summarizes our partition of ten cosmological tensions. The key distinction is between *frame artifacts* (tensions that arise from comparing measurements in incompatible H_0-dependent coordinate systems), *local-structure effects* (genuine astrophysical systematics in the nearby universe), and *physical residuals* (tensions that persist after both corrections and require modifications to the standard cosmological model).

3.2. *H_0 Tension: Frame Artifact and Local Structure*

The raw H_0 tension between SH0ES (73.04 ± 1.04 km s^{-1} Mpc^{-1}) and Planck (67.4 ± 0.5 km s^{-1} Mpc^{-1}) is 4.89σ. After ξ-normalization, $\sim$93% of this discrepancy is attributed to frame-mixing: the two datasets implicitly use different H_0 values in their distance definitions, producing an apparent tension that does not reflect a genuine disagreement about the expansion history.

The KBC void (Keenan et al. 2013) and bulk flow (Watkins et al. 2023) together account for an additional 64.5% of the residual: the KBC void produces $\Delta H_0 \approx 2.0$ km s^{-1} Mpc^{-1} (Kenworthy et al. 2019), and the bulk flow of $\sim$290 km s^{-1} contributes $\Delta H_0 \approx 1.6$ km s^{-1} Mpc^{-1} when sky-averaged at $z \sim 0.023$. The combined correction reduces the tension to 1.51σ, below the conventional discovery threshold.

Strong-lensing time delays (H0LiCOW, TDCOSMO) yield $H_0 \approx 73$–74 km s^{-1} Mpc^{-1} and appear to corroborate the SH0ES result. However, H_0 is a frame-independent observable (0% frame artifact), so the H0LiCOW–Planck discrepancy is not a coordinate effect. Rather, Birrer et al. (2021) demonstrated that when stellar kinematics are used to break the mass-sheet degeneracy (MSD), the inferred H_0 falls to $67.4^{+4.1}_{-3.2}$ km s^{-1} Mpc^{-1}, fully consistent with Planck at 0.0σ. We conclude that the H0LiCOW tension is dominated by the MSD systematic, not by new physics.

3.3. *Matter Density Deficit and S_8 Tension*

Table 1. Partition of Cosmological Tensions via ξ-Normalization

Tension	Probes	Frame artifact	Local structure	Physical residual	Status
H_0 (Cepheid)	SH0ES vs. Planck	∼93%	—	∼7% Ω_m	Resolved
H_0 (JWST)	JWST vs. Planck	∼93%	—	∼7% Ω_m	Resolved
BAO / Ω_m	DESI DR1	—	—	$\Omega_m = 0.295$	Physical
S_8 / σ_8	KiDS, DES, HSC	partial	—	2–3σ	Physical
Age tension	H_0 vs. stellar	dissolves	—	—	Resolved
A_L lensing	Planck	12–24%	—	statistical	Resolved
$w_0 w_a$	DESI DR1	—	—	$\Delta\chi^2 = 4.74$	Physical
H0LiCOW	Strong lensing	0%	0%	MSD†	Resolved
KBC void / flow	Local structure	0%	64.5%	1.51σ	Resolved
Ly-α BAO	BOSS/eBOSS $z = 2.33$	14%	—	86% (DE)	Physical

† Mass-sheet degeneracy; Birrer et al. (2021) MSD-corrected result is 0.0σ from Planck.

After removal of the frame artifact, the physical residual in H_0 corresponds to a matter density deficit: BAO χ^2 minimization over DESI DR1 yields $\Omega_m = 0.295 \pm 0.010$, compared to the Planck value of 0.315 ± 0.007 (2.2σ physical). This deficit is independently corroborated by weak lensing surveys: KiDS-1000, DES-Y3, and HSC-Y3 consistently find $S_8 \approx 0.763$–0.772, compared to the Planck prediction of $S_8 \approx 0.832$ (2.5–3σ). Because $S_8 = \sigma_8(\Omega_m/0.3)^{1/2}$, a 6% deficit in Ω_m alone forces S_8 downward by $\approx 3\%$, accounting for a significant fraction of the lensing tension without requiring modified gravity or suppressed structure growth.

3.4. *The $z = 0.51$ Anomaly and Dynamical Dark Energy*

The matter density deficit at $\Omega_m = 0.295$ reduces the global BAO χ^2 from 13.57 (Planck ΛCDM) to ∼11.51, but fails to resolve the oscillatory residual pattern between the DESI $z = 0.51$ and $z = 0.71$ bins. At $z = 0.51$, the observed $D_M/r_s = 13.62 \pm 0.25$ lies $+1.1\sigma$ above the $\Omega_m = 0.295$ prediction; at $z = 0.71$, the residual is negative. This sign flip is inconsistent with a pure Ω_m shift and instead requires a dynamical dark energy component.

Full $w_0 w_a$ minimization yields:

$$\Omega_m = 0.295 \pm 0.010 \tag{4}$$

$$w_0 = -0.634 \pm 0.09 \tag{5}$$

$$w_a = -1.388 \pm 0.35 \tag{6}$$

with $\Delta\chi^2 = 4.74$ relative to Ω_m-only LCDM (2.2σ preference, 2 additional degrees of freedom). The $w_a < 0$ "thawing" signature indicates that dark energy was less negative in the past and has become more so at late times.

3.5. *$O(1)$ Uncertainty Propagation via N/U Algebra*

The CPL prediction for $w(z = 2.33)$ is computed analytically using N/U Algebra (Martin 2026), in which each parameter is represented as a nominal–uncertainty pair (n, u) with conservative linear propagation rules:

$$(n_1, u_1) + (n_2, u_2) = (n_1 + n_2,\ u_1 + u_2), \tag{7}$$

$$k \cdot (n, u) = (k\,n,\ |k|\,u). \tag{8}$$

Applied to the DESI DR1 best-fit parameters ($w_0 = -0.634 \pm 0.090$, $w_a = -1.388 \pm 0.350$):

$$\begin{aligned} w(z{=}2.33) &= w_0 + \frac{z}{1+z}\,w_a \\ &= -0.6340 + 0.6997 \times (-1.388) \\ &= -1.605 \pm 0.335, \end{aligned} \tag{9}$$

where $z/(1+z) = 0.6997$ is a fixed scalar carrying no uncertainty. The N/U bound (± 0.335) is 1.28× wider than the Gaussian quadrature result (± 0.261), confirming the conservative property: the N/U interval contains the full Monte Carlo 1σ interval at $(-1.862, -1.341)$. The computation requires 2.85 μs, compared to 13.81 ms for a 10,000-sample Monte Carlo (4,844× faster).

The separation from ΛCDM ($w = -1.000$) is 1.81σ under N/U bounds. This is the honest propagated significance; the nominal gap of -0.605 without uncertainty propagation should not be quoted as a detection.

3.6. *Multi-Probe Dark Energy Convergence at $z > 2$*

To test whether the $w_0 w_a$ signal from DESI low-z BAO ($z = 0.3$–1.5) is consistent with the independent Ly-α BAO at $z = 2.33$, we fit $w_0 w_a$ to the six DESI bins at $z < 2.0$ only (excluding the Ly-α QSO bin) and then predict D_M/r_s at $z = 2.33$.

The low-z only best fit yields $w_0 = -0.578$, $w_a = -2.774$, and extrapolates to a D_M/r_s prediction of 38.83 at $z = 2.33$. The observed DESI Ly-α value is

39.71 ± 0.94, yielding a pull of $+0.94\sigma$ — within 1σ of the independent prediction.

The CPL equation of state at $z = 2.33$ from this extrapolation is:

$$w(z = 2.33) = w_0 + w_a \frac{z}{1+z} \approx -2.52 \qquad (10)$$

compared to the ΛCDM value of $w = -1.000$.

These two datasets — DESI low-z BAO and eBOSS Ly-α — were fit independently. Their convergence within 1σ is consistent with a coherent dark energy evolution signal spanning $z = 0.3$–2.3: the full redshift range of currently available BAO data. We note that this is consistent with but does not constitute a detection; confirmation requires the Euclid DR1 measurement (October 2026).

4. DISCUSSION

4.1. *What the Standard Model Gets Wrong (and Where)*

The partition in Table 1 carries a direct implication for the research agenda of the field. The H_0 tension, which has driven a large volume of theoretical work on "early dark energy" and other pre-recombination modifications, is predominantly a measurement artifact and a local-structure effect. Our analysis implies that the theoretical effort devoted to explaining the 5σ H_0 tension has been largely misdirected: the genuine physical signal is the Ω_m deficit and the $w_0 w_a$ dark energy evolution, not an anomalous expansion rate.

The Ly-α forest result reinforces this picture. The 86% physical fraction of the Ly-α BAO tension at $z = 2.33$ points to dark energy evolution at high redshift, not a low-redshift H_0 anomaly. This is a qualitatively different signal, and it requires a correspondingly different theoretical response.

4.2. *The UHA Framework as a Diagnostic*

The Universal Horizon Address (UHA; Martin 2026, Patent US 63/902,536) provides the coordinate infrastructure for this analysis. The ξ normalization defined in Equation (1) is the central operation: it strips H_0-dependent noise from observational comparisons, exposing the physical residuals. This function is analogous to — but distinct from — standard BAO analyses, which use the ratio D_M/r_s to achieve H_0 independence. The ξ approach is more general: it applies to distance ladder comparisons, lensing observables, and any measurement that depends on a comoving distance scale.

4.3. *Predictions for Euclid DR1 (October 2026)*

Our framework makes the following falsifiable predictions for Euclid DR1:

1. $S_8 \approx 0.808 \pm 0.017$ — consistent with the $\Omega_m = 0.295$ deficit and inconsistent with Planck ΛCDM.

2. The $w_0 w_a$ signal will be confirmed at $> 3\sigma$ when combined with DESI DR2 and Euclid weak lensing.

3. The Ly-α BAO at $z \sim 2.4$ (Euclid QSO sample) will show a D_H/r_s excess consistent with $w(z > 2) < -1$.

If these predictions are confirmed, the partition in Table 1 will stand as the reference framework for the transition from the "crisis in cosmology" narrative to a precision measurement of evolving dark energy.

5. CONCLUSION

We have applied ξ-normalized coordinates to partition ten cosmological tensions into coordinate-frame artifacts, local-structure effects, and physical residuals. The principal findings are:

1. ∼93% of the H_0 tension is a frame artifact. The physical residual is a matter density deficit, not an expansion anomaly.

2. The H0LiCOW strong-lensing tension is dominated by the mass-sheet degeneracy; when properly corrected, it is consistent with Planck at 0.0σ.

3. 64.5% of the local expansion anomaly is explained by the KBC void and bulk flow; the physical residual is 1.51σ (resolved).

4. The genuine physical signals are $\Omega_m = 0.295$ (matter deficit, confirmed by DESI DR1) and $w_0 = -0.634$, $w_a = -1.388$ (evolving dark energy, $\Delta\chi^2 = 4.74$).

5. A multi-probe convergence test finds that the DESI low-z $w_0 w_a$ fit predicts the Ly-α BAO at $z = 2.33$ within 0.94σ, consistent with a coherent dark energy evolution signal spanning $z = 0.3$–2.3.

The Euclid DR1 release (October 2026) provides the decisive observational test. The ξ framework, the tension partition, and the $w_0 w_a$ predictions reported here are pre-registered on Zenodo ahead of that release.

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